

1. A motorcyclist and his motorcycle have a combined mass of 480 kg.

The motorcyclist drives down a straight road that is inclined at an angle θ to the horizontal, where $\sin \theta = \frac{1}{12}$, with the engine of the motorcycle working at 3.5 kW. The motorcycle is moving at a constant speed of $V \text{ m s}^{-1}$. The resistance to the motion of the motorcycle is modelled as a constant force with magnitude $20V$ newtons.

Find the value of V .

(5)

2. A particle P of mass 1.5 kg moves under the action of a single force $\mathbf{F}$ newtons.

At time t seconds, $t \geq 0$, P has velocity $\mathbf{v}\text{ m s}^{-1}$, where

$$\mathbf{v} = (5t^2 - t^3)\mathbf{i} + (2t^3 - 8t)\mathbf{j}$$

- (a) Find $\mathbf{F}$ when $t = 2$

(4)

At time $t = 0$, P is at the origin O .

- (b) Find the position vector of P relative to O at the instant when P is moving in the direction of the vector $\mathbf{j}$

(4)

3.

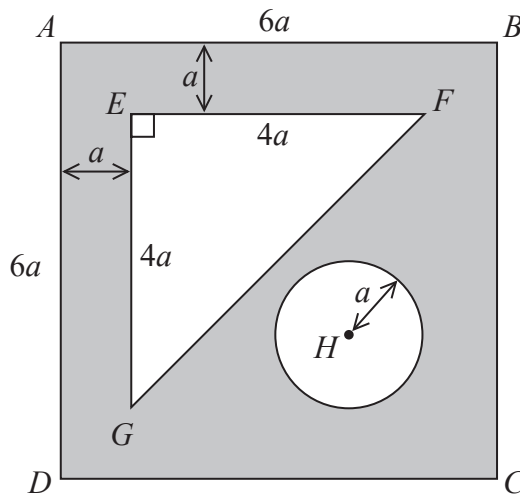


Figure 1

The uniform lamina $ABCD$ is a square of side $6a$. The template T , shown shaded in Figure 1, is formed by removing the right-angled triangle EFG and the circle, centre H and radius a , from the square lamina.

Triangle EFG has $EF = EG = 4a$, with EF parallel to AB and EG parallel to AD . The distance between AB and EF is a and the distance between AD and EG is a .

The point H lies on AC and the distance of H from BC is $2a$.

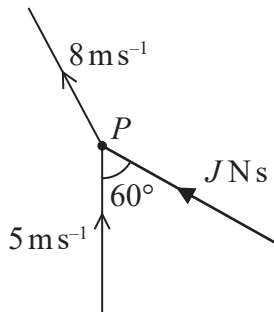
- (a) Show that the centre of mass of T is a distance $\frac{4(67 - 3\pi)}{3(28 - \pi)}a$ from AD . (5)

The template T is suspended from the ceiling by two light inextensible vertical strings. One string is attached to T at A and the other string is attached to T at B so that T hangs in equilibrium with AB horizontal.

The weight of T is W . The tension in the string attached to T at B is kW , where k is a constant.

- (b) Find the value of k , giving your answer to 2 decimal places. (3)

4.

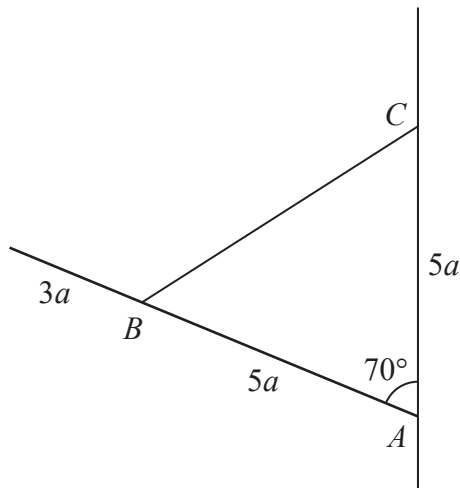
**Figure 2**

A particle P of mass 0.3 kg is moving with speed 5 m s^{-1} along a straight line on a smooth horizontal plane. The particle receives a horizontal impulse of magnitude $J \text{ N s}$. The speed of P immediately after receiving the impulse is 8 m s^{-1} . The angle between the direction of motion of P before it receives the impulse and the direction of the impulse is 60° , as shown in Figure 2.

Find the value of J .

(6)

5.

**Figure 3**

A uniform rod, of length $8a$ and mass M , has one end freely hinged to a fixed point A on a vertical wall. One end of a light inextensible string is attached to the rod at the point B , where $AB = 5a$. The other end of the string is attached to the wall at the point C , where $AC = 5a$ and C is vertically above A . The rod rests in equilibrium in a vertical plane perpendicular to the wall with angle $BAC = 70^\circ$, as shown in Figure 3.

(a) Find, in terms of M and g , the tension in the string.

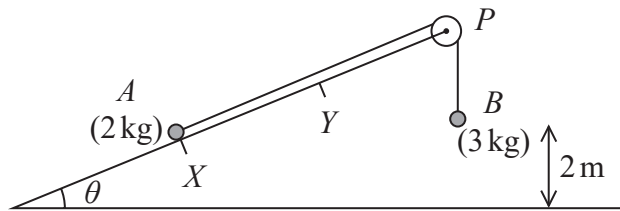
(3)

The magnitude of the force acting on the rod at A is λMg , where λ is a constant.

(b) Find, to 2 significant figures, the value of λ .

(6)

6.

**Figure 4**

Two particles, A and B , of mass 2 kg and 3 kg respectively, are connected by a light inextensible string. Particle A is held at rest at the point X on a fixed rough ramp that is inclined at an angle θ to the horizontal, where $\tan \theta = \frac{5}{12}$. The string passes over a small smooth pulley P that is fixed at the top of the ramp. Particle B hangs vertically below P , 2 m above the ground, as shown in Figure 4.

The particles are released from rest with the string taut so that A moves up the ramp and the section of the string from A to P is parallel to a line of greatest slope of the ramp.

The coefficient of friction between A and the ramp is $\frac{3}{8}$.

Air resistance is ignored.

(a) Find the potential energy lost by the system as A moves 2 m up the ramp.

(4)

(b) Find the work done against friction as A moves 2 m up the ramp.

(4)

When B hits the ground, B is brought to rest by the impact and does not rebound and A continues to move up the ramp.

(c) Use the work-energy principle to find the speed of B at the instant before it hits the ground.

(4)

Particle A comes to instantaneous rest at the point Y on the ramp, where $XY = (2 + d)\text{ m}$.

(d) Use the work-energy principle to find the value of d .

(4)

7. [In this question, the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are in a vertical plane, $\mathbf{i}$ being horizontal and $\mathbf{j}$ being vertically upwards.]

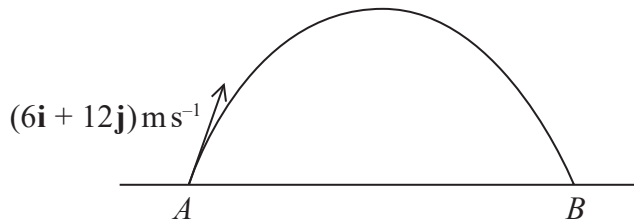


Figure 5

A small ball is projected with velocity $(6\mathbf{i} + 12\mathbf{j}) \text{ m s}^{-1}$ from a fixed point A on horizontal ground. The ball hits the ground at the point B , as shown in Figure 5. The motion of the ball is modelled as a particle moving freely under gravity.

- (a) Find the distance AB .

(4)

When the height of the ball above the ground is more than h metres, the speed of the ball is less than 10 m s^{-1}

- (b) Find the smallest possible value of h .

(4)

When the ball is at the point C on its path, the direction of motion of the ball is perpendicular to the direction of motion of the ball at the instant before it hits the ground at B .

- (c) Find, in terms of $\mathbf{i}$ and $\mathbf{j}$, the velocity of the ball when it is at C .

(3)

8. Particles A , B and C , of masses $2m$, m and $3m$ respectively, lie at rest in a straight line on a smooth horizontal plane with B between A and C . Particle A is projected towards particle B with speed $2u$ and collides directly with B .

The coefficient of restitution between each pair of particles is e .

- (a) (i) Show that the speed of B immediately after the collision with A is $\frac{4}{3}u(1+e)$

- (ii) Find the speed of A immediately after the collision with B .

(7)

At the instant when A collides with B , particle C is projected with speed u towards B so that B and C collide directly.

- (b) Show that there will be a second collision between A and B .

(6)
